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Qilin Xu

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Education

Carnegie Mellon University | School of Computer Science — *Aug. 2018*
M.S. in Information Technology — *Pittsburgh, PA*
Nanjing University | Kuang Yaming Honors School — *Jun. 2014*
B.S. in Oceanographic Science — *Nanjing, China*

Experience

LinkedIn, Senior Software Engineer — *Aug. 2022 - present*
AI Infra, Training Team — *Sunnyvale, CA*

- Collaborated closely with AI engineers to optimize LLM workflows, enabling faster training and inference at scale.
- Led the design of distributed LLM training optimizations, featuring *Triton*, *DeepSpeed, PyTorch DDP*, *etc.*
- Enabled cost-effective training for multi-billion and trillion-parameter models on existing hardware.
- Conducted company-wide demos and recurring knowledge-sharing sessions.

AI Infra, ML pipeline Team

- Founding member of *OpenConnect*, LinkedIn's next-gen ML pipeline powered by the open-source project ***Flyte***.
- As tech lead, brought key enhancements to *Flyte*, enabling flexible ML pipeline orchestration.
- Collaborated with AI product teams in productionizing LinkedIn's major ML workflows.

Hulu, Senior Software Engineer — *Dec. 2018 - Aug.2022*
ML & Data Infra Team — *Santa Monica, CA*

- Led improvements to the company's major training pipelines with *Accelerator.*
- Member of the data center migration squad, managing AWS infrastructure migration with Kubernetes.

Payment Lifecycle Team

- Led *Hulu* and *Disney+* payment platform unification, bringing millions of subscribers since launch.

Project

***Liger* - LinkedIn's GPU-Efficient Runtime for LLM training**
Core Contributor

- Developed Triton-fused GPU kernels with efficient training techniques (multipacking, PEFT, etc.).
- Authored key LLM layers (*RMSNorm*, *RoPE etc.)* with improved GPU utilization and memory savings.
- Contributed to *AutoTune*, a feature that dynamically generates optimal training configurations based on model parameters, GPU resources, sequence length, and other factors.

***OpenConnect/Flyte(open-sourced)* - LinkedIn's Next-Generation ML Pipeline**
Core Contributor

- Spearheaded the design of *OpenConnect*, enhancing LinkedIn's model training, deployment, and monitoring.
- Extended *Flyte*'s core functionality with customized features tailored to LinkedIn's AI needs, including workflow and task lifecycle management, cache control, and more.
- Built and maintained major model training components, including model publisher, calibration, etc.
- Facilitated the adoption of *Flyte* across the organization with scalable, reliable ML workflows.

***Martel* - LinkedIn's inhouse Deep Learning library**
Owner

- Architected LinkedIn's Deep Learning library focusing on distributed data loading capabilities.
- Integrated robust support for *PyTorch DDP* to enable efficient multi-node, multi-GPU training.
- Provided efficient conversions between LinkedIn's feature store and native sparse/dense tensors.

Graph Neural Network (GNN) Offline Training
Core Contributor

- Architected Kubernetes resource orchestration for LinkedIn's GNN graph engine.
- Optimized resource allocation and scalability for large-scale graph processing.

Interactive Dev
Owner

- Built a cloud-native solution that supports direct model development within pods, facilitating rapid prototyping.
- Achieved significant reduction in iteration time from hours to seconds, enabling fast experimentation cycles.

Professional Skills

Platforms/Programming: Java, Python, Golang, Kotlin, C++, Kubernetes, Docker, Kafka, Spark, AWS, Flink, Hadoop
AI Infra: Flyte, PyTorch, Triton, Keras, TensorFlow, Ray, Scikit-learn, XGBoost